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Design and Implementation of 4-bit Signed Multiplier Using Verilog

Open Ended Laboratory Experiment – Verilog HDL using Vivado

1. Introduction:

The objective of this experiment is to design, simulate, and implement a 4-bit signed binary multiplier using Verilog Hardware Description Language (HDL). The design is developed and verified using Vivado. A 4-bit signed multiplier accepts two 4-bit binary inputs and produces an 8-bit output representing the product of the inputs. Since the operands are signed, both positive and negative values are considered.
This experiment helps in understanding combinational logic design, binary arithmetic operations, HDL-based modeling, simulation, synthesis, and hardware implementation on FPGA platforms.

2.Aim of Experiment:

- To design and implement a 4-bit signed multiplier
- To verify using stimulation
- To implement design on Atrix-7 FPGA

3. Problem statement:

Design and implement a 4-bit signed multiplier usingVerilogHDL that accepts two 4-bit signed inputs and generatesan8-bit signed output. The design should be suitable for FPGAimplementation, simulated for correctness, and verified through hardware implementation.
For signed numbers like:
i)Positive & positive numbers

A = 0101 → +5
B = 0011 → +3

P = 00001111 → +15

ii)Positive & Negative numbers:

A = 0011 → +3
B = 1110 → -2 (2's complement)

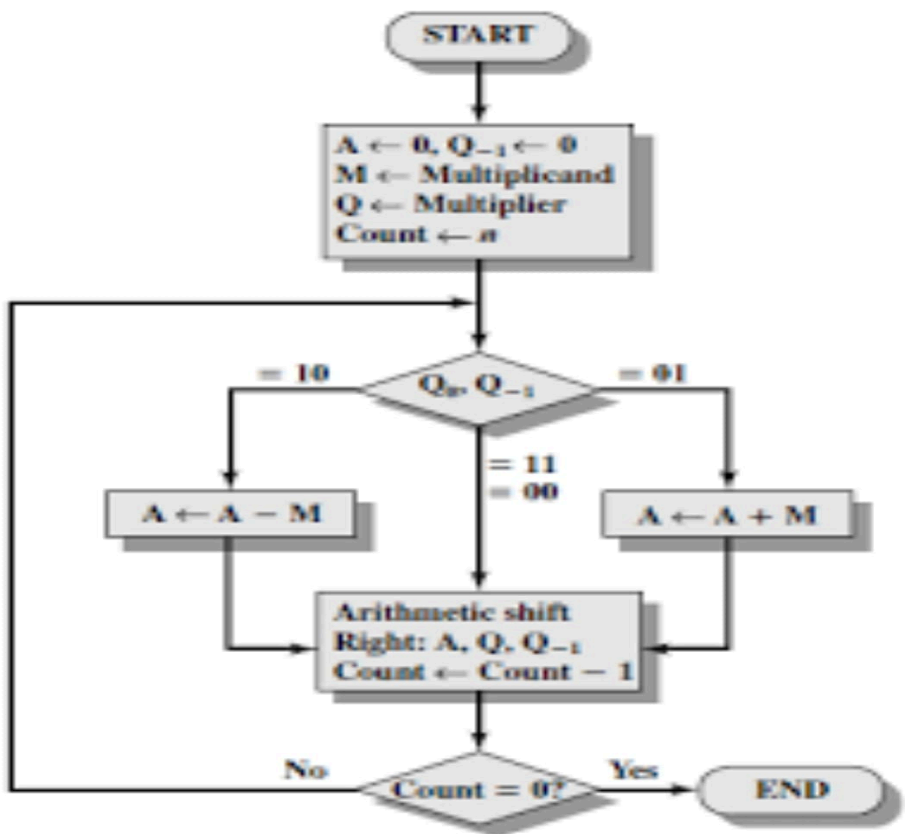
P = 11111010 → -6

iii) Negative & Negative numbers:

A = 1100 → -4 (2's complement)
B = 1101 → -3 (2's complement)

P = 00001100 → +12

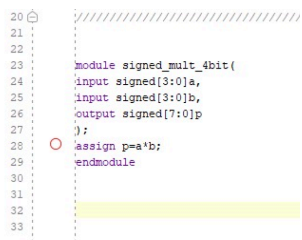
4. Block Diagram / Flow Chart:



5.Verilog Code and Testbench:

5.1 Verilog Code for 4-bit Signed Multiplier :

```
module signed_multiplier_4bit (
input signed [3:0] A,
input signed [3:0] B,
output signed [7:0] P );
assign P = A * B;
endmodule
```

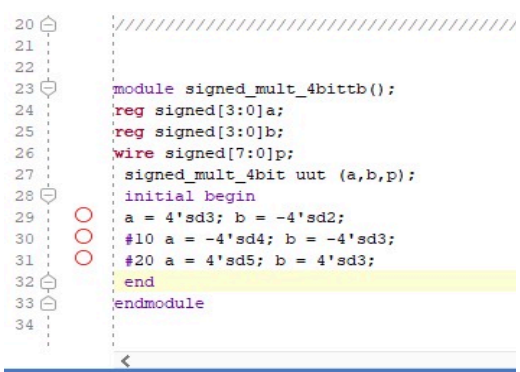


```
/////////////////////////////////

module signed_mult_4bit(
input signed[3:0]a,
input signed[3:0]b,
output signed[7:0]p
);
assign p=a*b;
endmodule
```

5.2 Testbench Code :

```
module signed_multiplier_4bittb();
reg signed [3:0] A,
reg signed [3:0] B,
wire signed [7:0] P
signed_multiplier_4bit uut (A, B , P);
initial begin A = 4'sd3; B = -4'sd2; // -6
#10 A = -4'sd4; B = -4'sd3; // +12
#10 A = 4'sd5; B = 4'sd3; // +15
end
endmodule
```



```
////////////////////////////////////////

module signed_mult_4bittb();
reg signed[3:0]a;
reg signed[3:0]b;
wire signed[7:0]p;
 signed_mult_4bit uut (a,b,p);
 initial begin
 a = 4'sd3; b = -4'sd2;
 #10 a = -4'sd4; b = -4'sd3;
 #20 a = 4'sd5; b = 4'sd3;
 end
endmodule
```

6. Simulation Results:

The design was simulated using the Vivado behavioral simulation tool. The testbench applied various input combinations to verify correct multiplication operation. The observed output matched the expected results for all test cases.

Example Results:

1st case:

A = 0101 → +5
B = 0011 → +3

P = 00001111 → +15

2nd case:

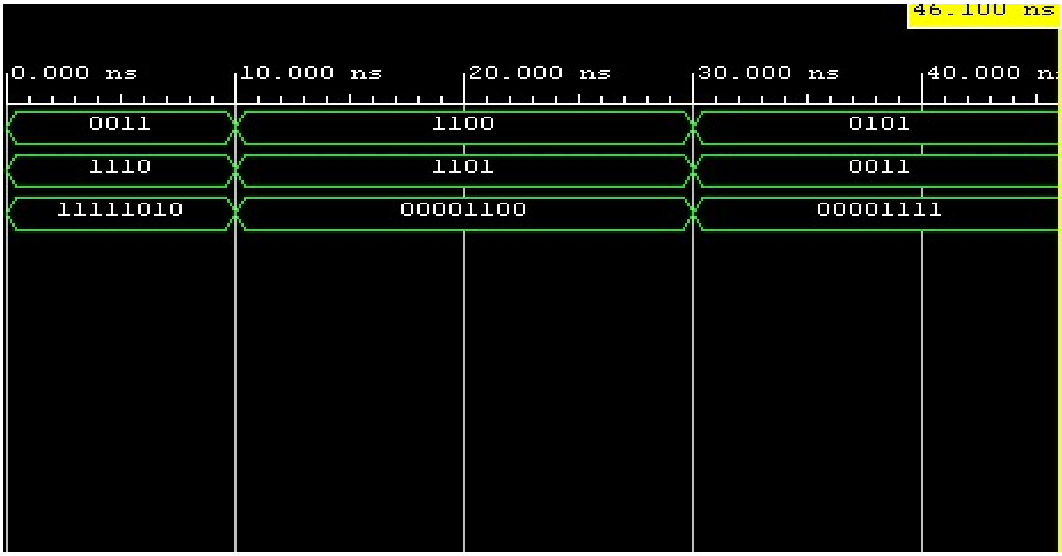
A = 0011 → +3
B = 1110 → -2 (2's complement)

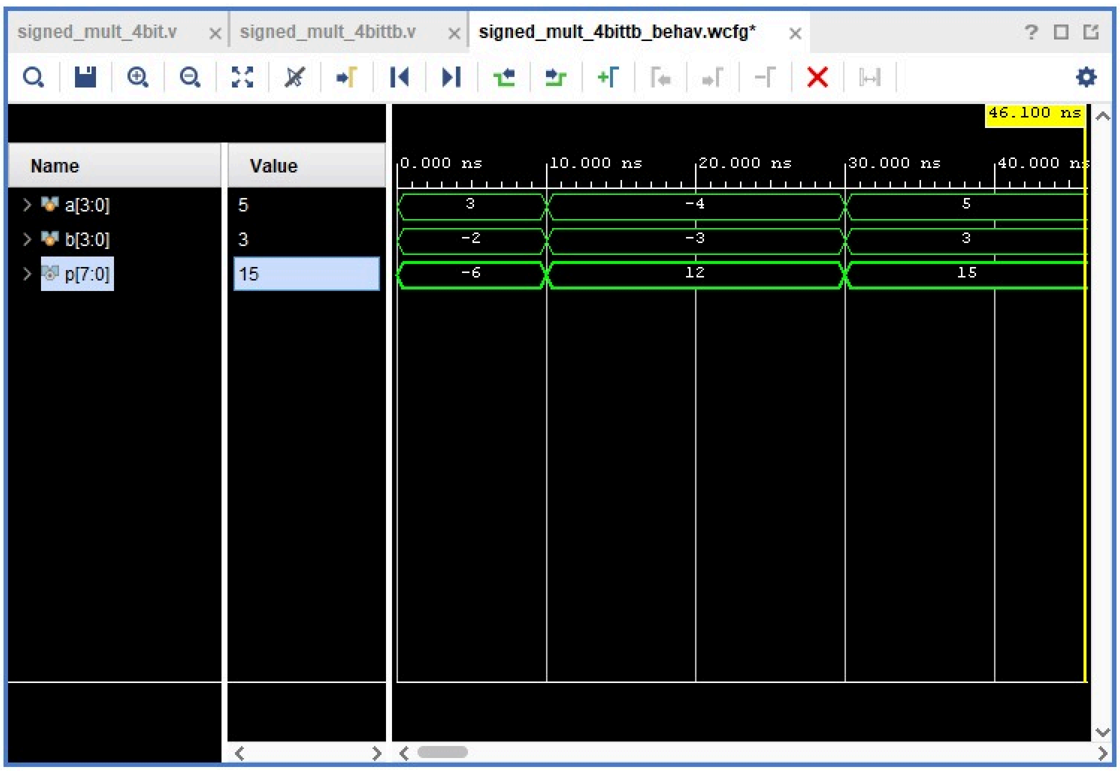
P = 11111010 → -6

3rd case:

A = 1100 → -4 (2's complement)
B = 1101 → -3 (2's complement)

P = 00001100 → +12

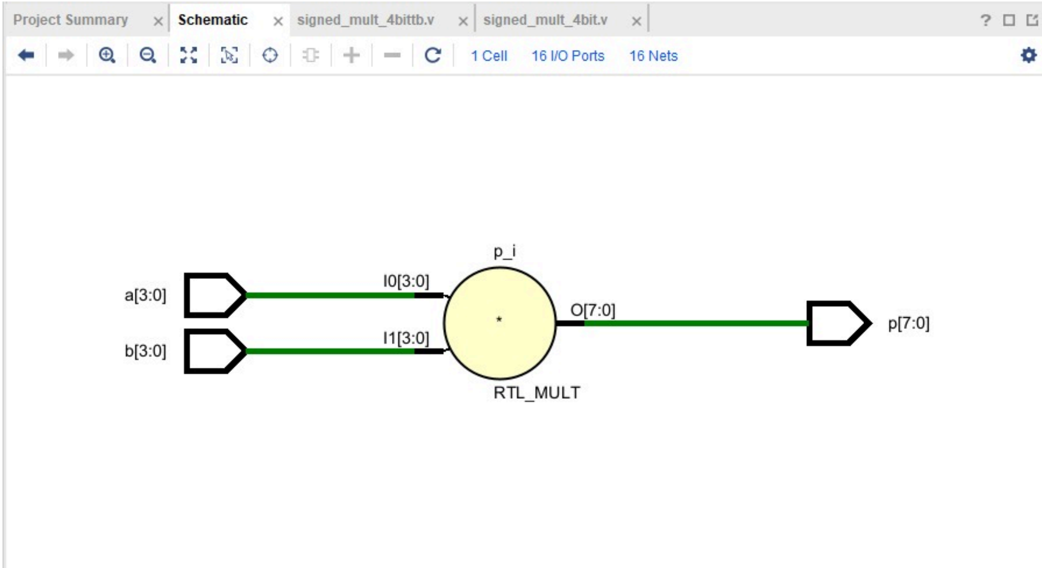




7. Implementation:

After successful simulation, the design was synthesized and implemented using Vivado. The Verilog code was translated into FPGA hardware resources such as LUTs and combinational logic. The design does not use any clock signal and is purely combinational.

7.1. SCHEMATIC:



7.2. Pin configuration:

Name	Direction	Interface	Neg Diff Pair	Package Pin	Fixed	Bank	I/O Std	Vcco	Vref	Drive Strength	Slew Type	Pull Type	Off-Chip Terminatior
a[3]	IN			D9	✓	16	LVCMOS33*	3.300				NONE	NONE
a[2]	IN			C9	✓	16	LVCMOS33*	3.300				NONE	NONE
a[1]	IN			B9	✓	16	LVCMOS33*	3.300				NONE	NONE
a[0]	IN			B8	✓	16	LVCMOS33*	3.300				NONE	NONE
b[3]	IN			A8	✓	16	LVCMOS33*	3.300				NONE	NONE
b[2]	IN			C11	✓	16	LVCMOS33*	3.300				NONE	NONE
b[1]	IN			C10	✓	16	LVCMOS33*	3.300				NONE	NONE
b[0]	IN			A10	✓	16	LVCMOS33*	3.300				NONE	NONE
p[7]	OUT			H5	✓	35	LVCMOS33*	3.300		12		NONE	FP_VTT_50
p[6]	OUT			J5	✓	35	LVCMOS33*	3.300		12		NONE	FP_VTT_50
p[5]	OUT			T9	✓	14	LVCMOS33*	3.300		12		NONE	FP_VTT_50
p[4]	OUT			T10	✓	14	LVCMOS33*	3.300		12		NONE	FP_VTT_50
p[3]	OUT			G6	✓	35	LVCMOS33*	3.300		12		NONE	FP_VTT_50
p[2]	OUT			F6	✓	35	LVCMOS33*	3.300		12		NONE	FP_VTT_50
p[1]	OUT			E1	✓	35	LVCMOS33*	3.300		12		NONE	FP_VTT_50
p[0]	OUT			G1	✓	35	LVCMOS33*	3.300		12		NONE	FP_VTT_50

8. Synthesis Reports:

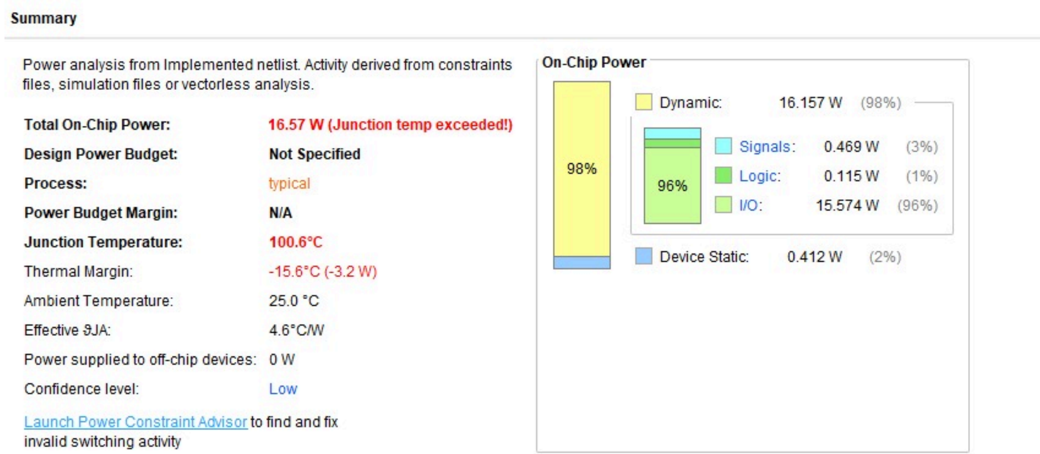
8.1 Timing Analysis:

Design Timing Summary

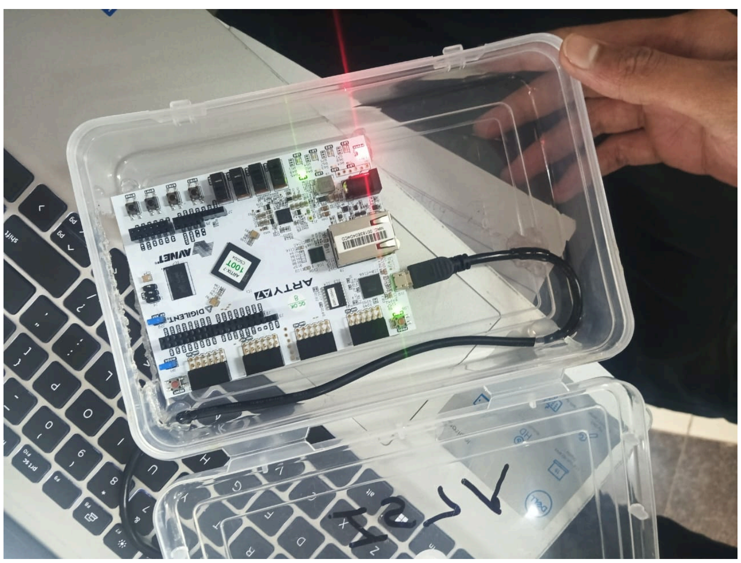
Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	inf	Worst Hold Slack (WHS):	inf	Worst Pulse Width Slack (WPWS):	NA
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	NA
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	NA
Total Number of Endpoints:	8	Total Number of Endpoints:	8	Total Number of Endpoints:	NA

There are no user specified timing constraints.

8.2 Power Analysis:



8.2 OutPut:



9. Applications:

- Arithmetic Logic Units (ALUs)
- Digital signal processing systems
- Microprocessors and controllers
- FPGA-based arithmetic circuits

CONCLUSION:

In this experiment, a 4-bit signed multiplier was successfully designed, simulated, and implemented using Verilog HDL and Vivado. The simulation and synthesis results confirm the correctness and efficiency of the design. This experiment provides practical insight into digital arithmetic design and FPGA-based implementation.